

Question:

What is the order of the element 5 in U_8 ?

Answer Choices:

- A. 3
- B. 4
- C. 1
- D. 5
- E. 9
- F. 2
- G. 8
- H. 0
- I. 6
- J. 7

Reasoning + Answer:

The question asks for the order of the element 5 in the group U_8 .

First, understand what U_8 is:

U_n denotes the multiplicative group of units modulo n , i.e., the set of integers between 1 and $n-1$ that are coprime to n , under multiplication modulo n .

So, U_8 is the multiplicative group of units modulo 8.

Step 1: List the elements of U_8

Numbers from 1 to 7 that are coprime to 8:

8 is 2^3 , so numbers coprime to 8 are those not divisible by 2.

Thus:
1, 3, 5, 7

So $U_8 = \{1, 3, 5, 7\}$

Step 2: Find the order of the element 5 in this group.

The order of an element a in a group is the smallest positive integer k such that $a^k \equiv 1 \pmod 8$.

We compute powers of 5 modulo 8:

$5^1 \pmod 8 = 5$
 $5^2 = 25 \pmod 8 = 25 - 3 \cdot 8 = 25 - 24 = 1$

So $5^2 \equiv 1 \pmod 8$

Therefore, the order is 2.

Answer: The order of 5 in U_8 is 2.

<answer> F </answer>